



The French National Research Institute for Agriculture, Food and the Environment (INRAE) is a public research institution bringing together a working community of 12,000 people, with more than 200 research units and 42 experimental units located throughout France. INRAE ranks among the world's leading institutions in agricultural and food sciences, plant sciences and animal sciences. Its research aims to develop solutions for multi-performance agriculture, high-quality food and sustainable management of resources and ecosystems.

OUTLINE OF THE SCIENTIFIC PROPOSAL & ACTIVITIES

■ In the context of the Horizon Europe DOMINO project (<https://www.domino-euproject.eu/>) coordinated by Stéphane Chaillou leader of the Food Microbial Ecology research group at Micalis Institute of INRAE (<https://fme.micalis.fr/>), Jouy-en-Josas, we are opening a 12-to-18 months postdoctoral position to work on food microbiome engineering. More precisely, the project aims at developing synthetic ecology approaches and metabolic modelling of Kimchi-type fermented foods.

■ Kimchi microbiome engineering, a top-down approach. In the first two years of the DOMINO project (2023-2025), a large collection of bacterial strains from traditional, spontaneous Kimchi fermentations were collected and characterized at the genomic level by the two partners (TFTAK and INRAE). This large microbial resource can now be used to design tailor-made consortia and to study the role and activities of these consortia on Kimchi fermentation. In addition, a liquid-based kimchi-like growth media has been developed to serve as a simplified matrix or substrate for high-throughput screening of consortia. During the first step of the synthetic ecology approach, a bottom-up strategy was used starting from genome-scale modelling to design few microbial consortia which could represent a minimal community showing potential metabolic interactions. These tailor-made consortia are now being tested in real Kimchi fermentation.

■ The postdoctoral researcher will have in charge to carry out, a top-down strategy often used as alternative in synthetic ecology approaches and to compare the results with those obtained from the bottom-up strategy. The project has three objectives:

The first objective of the postdoctoral project will be to perform random combinations of various strains using factorial design and screening algorithms such as Random Forest algorithm to establish how many different gene expression profiles (different functional behavior of the microbial communities) can be determined during Kimchi fermentation. For this purpose, high throughput fermentations in micro-fermenters will be performed and a new omics analytical tool (Food Microbiome Metabolic Modules or F3M suite, developed also in the frame of the DOMINO project, ([DOI](#))) will be applied to study the metabolic interactions and gene expression profiles at several taxonomic and/or functional levels. The purpose of this strategy is to select one or two consortia representative of each functional pattern.

The second objective is to study how environmental parameters such as temperature, oxygen and the degree of solidity of the fermented food has an impact on the functional and structural pattern of the selected consortia. A collaboration with another MICALIS team which will be equipped in 2026 with a food 3D printer will enable controlled experimental implementation of the spatial parameter (semi-solid to solid) of Kimchi models. At last, results will be compared to those obtained in real kimchi fermentations and to those obtained by the bottom-up strategy.

The third objective will be to study how the selected kimchi designed consortia interact metabolically with various synthetic gut microbial communities (designed to match the different main gut microbiome states).

YOUR PROFILE & GENERAL INFORMATION

 A dynamic research environment, benefit from interdisciplinary collaborations, and receive supportive mentoring!

 Location : Micalis Institute (<https://www.micalis.fr/>), INRAE centre of Jouy-en-Josas (30km south west of Paris) - University of Paris-Saclay.

 Start date : April 2026 (flexible). **Deadline for application end of January 2023.**

Duration: 12-18 months depending on post PhD professional experiences.

 Candidate profile :

- PhD in microbial ecology and/or microbial metagenomic or related field
- Good background in bioinformatics, biostatistics is recommended.
- A skill in metabolic networks modelling would be an asset.
- Interest in interdisciplinary work and integration of biological data.

- **Important: Micalis Institute is a ZRR (Zone à Régime Restrictif) and the selected candidate will have to process with security defense check and authorization. A delay of 4 months is expected between the selection and recruitment.**

 Contact to apply for the position : Dr. Stéphane CHAILLOU - stephane.chaillou@inrae.fr
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YOUR QUALITY OF LIFE AT INRAE

By joining INRAE, depending on the type of contract, you will be entitled to:

- up to 30 days of holiday + 15 days of RTT (reduced working time) per year (for full-time employees)
- parenting support: CESU childcare vouchers, leisure benefits;
- skills development programs: training, career guidance;
- social support: advice and counselling, social assistance and loans;
- holiday and leisure benefits: holiday vouchers, accommodation at preferential rates;
- sports and cultural activities;
- staff catering.